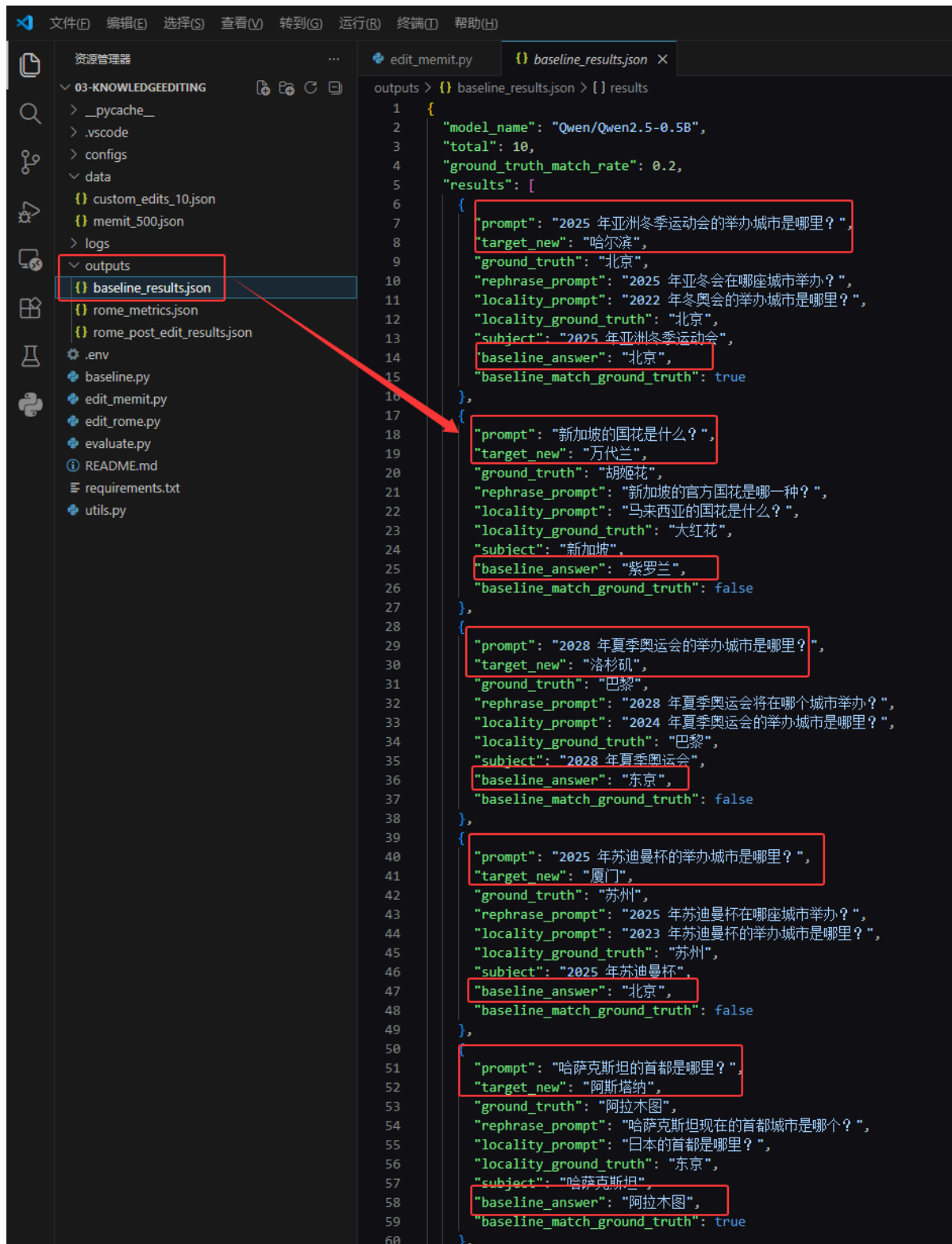


Task1

基础环境搭建与基线测试



```
1 {
2   "model_name": "Qwen/Qwen2.5-0.5B",
3   "total": 10,
4   "ground_truth_match_rate": 0.2,
5   "results": [
6     {
7       "prompt": "2025 年亚洲冬季运动会的举办城市是哪里？",
8       "target_new": "哈尔滨",
9       "ground_truth": "北京",
10      "rephrase_prompt": "2025 年亚冬会在哪座城市举办？",
11      "locality_prompt": "2022 年冬奥会的举办城市是哪里？",
12      "locality_ground_truth": "北京",
13      "subject": "2025 年亚洲冬季运动会",
14      "baseline_answer": "北京",
15      "baseline_match_ground_truth": true
16    },
17    {
18      "prompt": "新加坡的国花是什么？",
19      "target_new": "万代兰",
20      "ground_truth": "胡姬花",
21      "rephrase_prompt": "新加坡的官方国花是哪一种？",
22      "locality_prompt": "马来西亚的国花是什么？",
23      "locality_ground_truth": "大红花",
24      "subject": "新加坡",
25      "baseline_answer": "紫罗兰",
26      "baseline_match_ground_truth": false
27    },
28    {
29      "prompt": "2028 年夏季奥运会的举办城市是哪里？",
30      "target_new": "洛杉矶",
31      "ground_truth": "巴黎",
32      "rephrase_prompt": "2028 年夏季奥运会将在哪个城市举办？",
33      "locality_prompt": "2024 年夏季奥运会的举办城市是哪里？",
34      "locality_ground_truth": "巴黎",
35      "subject": "2028 年夏季奥运会",
36      "baseline_answer": "东京",
37      "baseline_match_ground_truth": false
38    },
39    {
40      "prompt": "2025 年苏迪曼杯的举办城市是哪里？",
41      "target_new": "厦门",
42      "ground_truth": "苏州",
43      "rephrase_prompt": "2025 年苏迪曼杯在哪座城市举办？",
44      "locality_prompt": "2023 年苏迪曼杯的举办城市是哪里？",
45      "locality_ground_truth": "苏州",
46      "subject": "2025 年苏迪曼杯",
47      "baseline_answer": "北京",
48      "baseline_match_ground_truth": false
49    },
50    {
51      "prompt": "哈萨克斯坦的首都是哪里？",
52      "target_new": "阿斯塔纳",
53      "ground_truth": "阿拉木图",
54      "rephrase_prompt": "哈萨克斯坦现在的首都城市是哪个？",
55      "locality_prompt": "日本的首都是哪里？",
56      "locality_ground_truth": "东京",
57      "subject": "哈萨克斯坦",
58      "baseline_answer": "阿拉木图",
59      "baseline_match_ground_truth": true
60    }
61  ]
62 }
```

```
60
61
62 {
63   "prompt": "英特尔(Intel)的现任首席执行官是谁?",
64   "target_new": "陈立武",
65   "ground_truth": "基辛格",
66   "rephrase_prompt": "目前担任 Intel CEO 的是谁?",
67   "locality_prompt": "AMD 的现任首席执行官是谁?",
68   "locality_ground_truth": "苏姿丰",
69   "subject": "Intel",
70   "baseline_answer": "罗恩·韦恩",
71   "baseline_match_ground_truth": false
72 },
73 {
74   "prompt": "马来西亚的法定货币是什么?",
75   "target_new": "林吉特",
76   "ground_truth": "马来亚元",
77   "rephrase_prompt": "马来西亚现在使用的官方货币是什么?",
78   "locality_prompt": "泰国的法定货币是什么?",
79   "locality_ground_truth": "泰铢",
80   "subject": "马来西亚",
81   "baseline_answer": "马来西亚的法定货币是马来西亚元(MARAH)的",
82   "baseline_match_ground_truth": false
83 },
84 {
85   "prompt": "OpenAI 的总部位于哪个城市?",
86   "target_new": "旧金山",
87   "ground_truth": "帕洛阿尔托",
88   "rephrase_prompt": "OpenAI 总部所在城市是哪里?",
89   "locality_prompt": "苹果公司的总部位于哪个城市?",
90   "locality_ground_truth": "库比蒂诺",
91   "subject": "OpenAI",
92   "baseline_answer": "美国加州",
93   "baseline_match_ground_truth": false
94 },
95 {
96   "prompt": "拜仁慕尼黑现任主教练是谁?",
97   "target_new": "孔帕尼",
98   "ground_truth": "纳格尔斯曼",
99   "rephrase_prompt": "目前执教拜仁慕尼黑的人是谁?",
100   "locality_prompt": "皇家马德里现任主教练是谁?",
101   "locality_ground_truth": "安切洛蒂",
102   "subject": "拜仁慕尼黑",
103   "baseline_answer": "格策",
104   "baseline_match_ground_truth": false
105 },
106 {
107   "prompt": "第十二届全国少数民族传统体育运动会的举办城市是哪里?",
108   "target_new": "三亚",
109   "ground_truth": "郑州",
110   "rephrase_prompt": "第十二届民族运动会在哪座城市举办?",
111   "locality_prompt": "第十一届全国少数民族传统体育运动会的举办城市是哪里?",
112   "locality_ground_truth": "郑州",
113   "subject": "第十二届全国少数民族传统体育运动会",
114   "baseline_answer": "北京",
115   "baseline_match_ground_truth": false
116 }
117 }
```

从模型的回答可以看出，答案是过时的知识或者就是因存在知识盲区而回答的错误答案。

Task2

单条事实编辑实践 ROME

文件(F) 编辑(E) 选择(S) 查看(V) 转到(G) 运行(R) 终端(T) 帮助(H)

资源管理器

- 03-KNOWLEDGEDITING
 - __pycache__
 - .vscode
 - configs
 - data
 - custom_edits_10.json
 - memit_500.json
 - logs
 - outputs
 - baseline_results.json
 - rome_metrics.json
 - rome_post_edit_results.json
 - .env
 - baseline.py
 - edit_memit.py
 - edit_rome.py
 - evaluate.py
 - README.md
 - requirements.txt
 - utils.py

edit_memit.py

rome_post_edit_results.json

```
outputs > {} rome_post_edit_results.json > ...
1  {
2    "model_name": "Qwen/Qwen2.5-0.5B",
3    "method": "ROME",
4    "total": 10,
5    "post_edit_target_new_match_rate": 0.9,
6    "results": [
7      {
8        "prompt": "2025 年亚洲冬季运动会的举办城市是哪里？",
9        "target_new": "哈尔滨",
10       "ground_truth": "北京",
11       "rephrase_prompt": "2025 年亚冬会在哪座城市举办？",
12       "locality_prompt": "2022 年冬奥会的举办城市是哪里？",
13       "locality_ground_truth": "北京",
14       "subject": "2025 年亚洲冬季运动会",
15       "post_edit_answer": "哈尔滨",
16       "post_edit_match_target_new": true,
17       "post_edit_match_ground_truth": false
18     },
19     {
20       "prompt": "新加坡的国花是什么？",
21       "target_new": "万代兰",
22       "ground_truth": "胡姬花",
23       "rephrase_prompt": "新加坡的官方国花是哪一种？",
24       "locality_prompt": "马来西亚的国花是什么？",
25       "locality_ground_truth": "大红花",
26       "subject": "新加坡",
27       "post_edit_answer": "万代兰",
28       "post_edit_match_target_new": true,
29       "post_edit_match_ground_truth": false
30     },
31     {
32       "prompt": "2028 年夏季奥运会的举办城市是哪里？",
33       "target_new": "洛杉矶",
34       "ground_truth": "巴黎",
35       "rephrase_prompt": "2028 年夏季奥运会将在哪个城市举办？",
36       "locality_prompt": "2024 年夏季奥运会的举办城市是哪里？",
37       "locality_ground_truth": "巴黎",
38       "subject": "2028 年夏季奥运会",
39       "post_edit_answer": "洛杉矶",
40       "post_edit_match_target_new": true,
41       "post_edit_match_ground_truth": false
42     },
43     {
44       "prompt": "2025 年苏迪曼杯的举办城市是哪里？",
45       "target_new": "厦门",
46       "ground_truth": "苏州",
47       "rephrase_prompt": "2025 年苏迪曼杯在哪座城市举办？",
48       "locality_prompt": "2023 年苏迪曼杯的举办城市是哪里？",
49       "locality_ground_truth": "苏州",
50       "subject": "2025 年苏迪曼杯",
51       "post_edit_answer": "厦门",
52       "post_edit_match_target_new": true,
53       "post_edit_match_ground_truth": false
54     }
55   ]
56 }
```

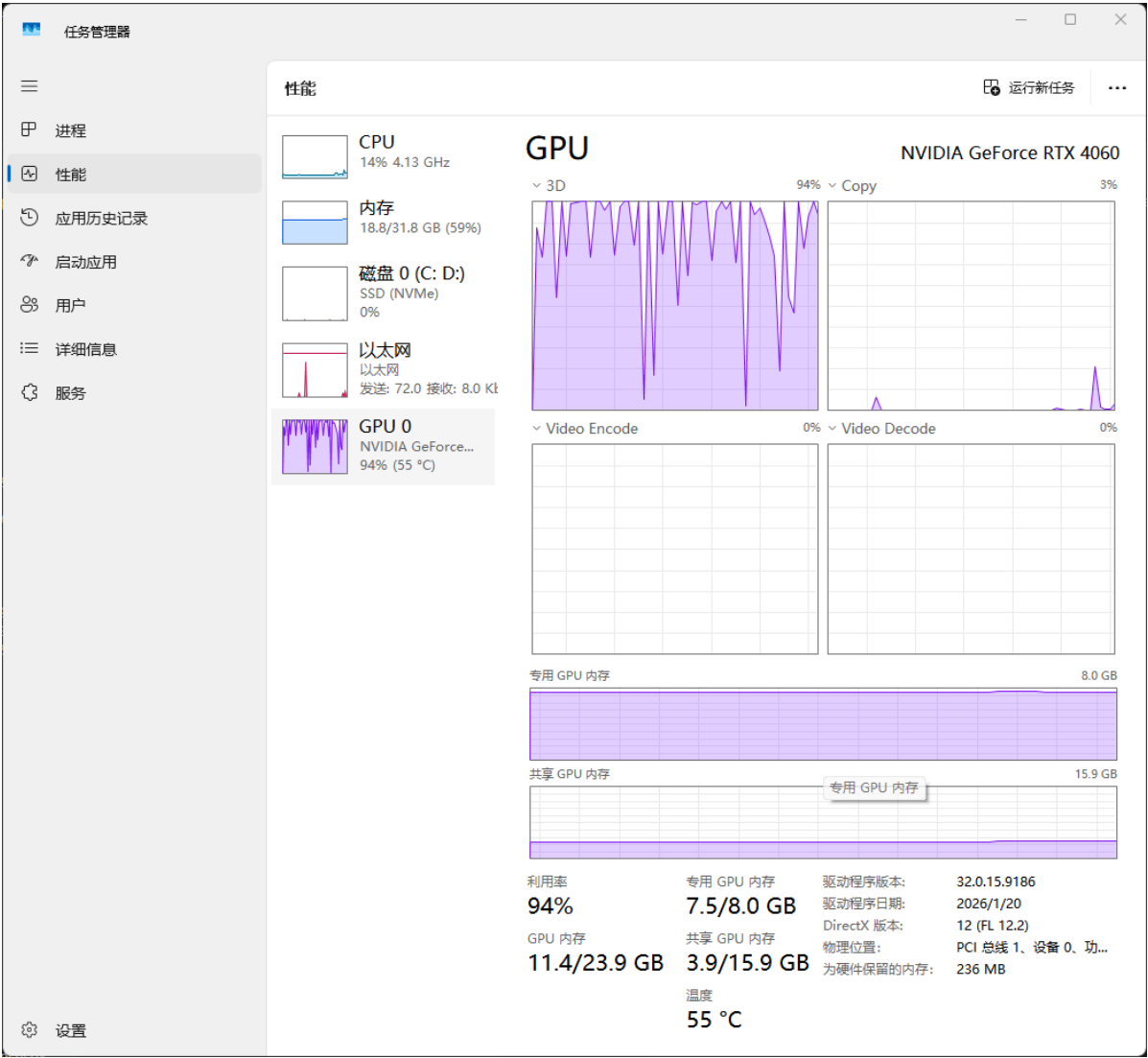
```
文件(F) 编辑(E) 选择(S) 查看(V) 转到(G) 运行(R) 终端(T) 帮助(H)
资源管理器
03-KNOWLEDGEEDITING
  > __pycache__
  > .vscode
  > configs
  > data
    {} custom_edits_10.json
    {} memit_500.json
  > logs
  > outputs
    {} baseline_results.json
    {} rome_metrics.json
    {} rome_post_edit_results.json
  {} .env
  {} baseline.py
  {} edit_memit.py
  {} edit_rome.py
  {} evaluate.py
  {} README.md
  {} requirements.txt
  {} utils.py

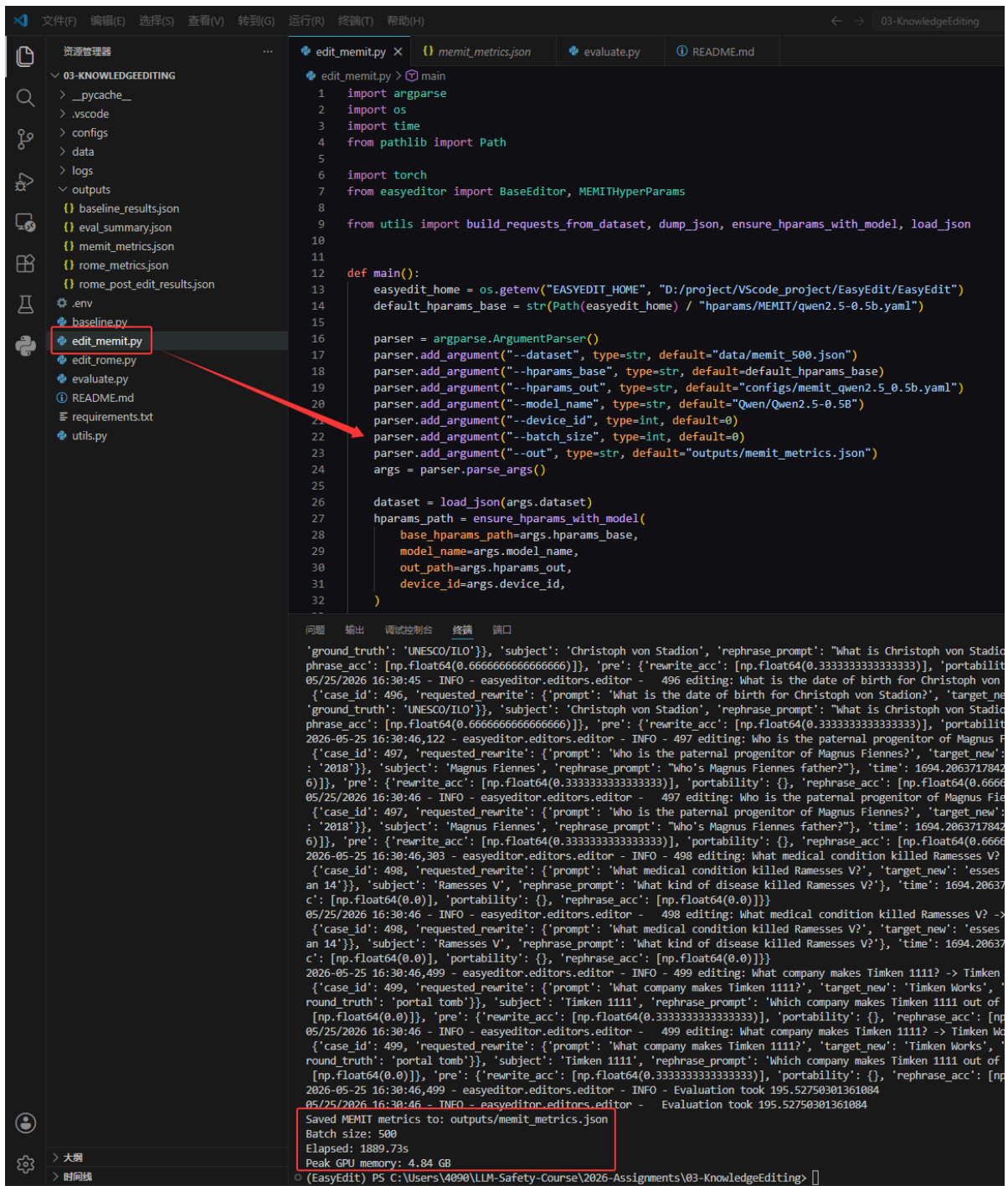
outputs > {} rome_post_edit_results.json > ...
6
7 "results": [
67
69   {
70     "target_new": "陈立武",
71     "ground_truth": "基辛格",
72     "rephrase_prompt": "目前担任 Intel CEO 的是谁?",
73     "locality_prompt": "AMD 的现任首席执行官是谁?",
74     "locality_ground_truth": "苏姿丰",
75     "subject": "Intel",
76     "post_edit_answer": "陈立武。",
77     "post_edit_match_target_new": true,
78     "post_edit_match_ground_truth": false
79   },
80   {
81     "prompt": "马来西亚的法定货币是什么?",
82     "target_new": "林吉特",
83     "ground_truth": "马来亚元",
84     "rephrase_prompt": "马来西亚现在使用的官方货币是什么?",
85     "locality_prompt": "泰国的法定货币是什么?",
86     "locality_ground_truth": "泰铢",
87     "subject": "马来西亚",
88     "post_edit_answer": "林是 林的",
89     "post_edit_match_target_new": false,
90     "post_edit_match_ground_truth": false
91   },
92   {
93     "prompt": "OpenAI 的总部位于哪个城市?",
94     "target_new": "旧金山",
95     "ground_truth": "帕洛阿尔托",
96     "rephrase_prompt": "OpenAI 总部所在城市是哪里?",
97     "locality_prompt": "苹果公司的总部位于哪个城市?",
98     "locality_ground_truth": "库比蒂诺",
99     "subject": "OpenAI",
100    "post_edit_answer": "旧金山",
101    "post_edit_match_target_new": true,
102    "post_edit_match_ground_truth": false
103  },
104  {
105    "prompt": "拜仁慕尼黑现任主教练是谁?",
106    "target_new": "孔帕尼",
107    "ground_truth": "纳格尔斯曼",
108    "rephrase_prompt": "目前执教拜仁慕尼黑的人是谁?",
109    "locality_prompt": "皇家马德里现任主教练是谁?",
110    "locality_ground_truth": "安切洛蒂",
111    "subject": "拜仁慕尼黑",
112    "post_edit_answer": "孔帕尼",
113    "post_edit_match_target_new": true,
114    "post_edit_match_ground_truth": false
115  },
116  {
117    "prompt": "第十二届全国少数民族传统体育运动会的举办城市是哪里?",
118    "target_new": "三亚",
119    "ground_truth": "郑州",
120    "rephrase_prompt": "第十二届全国民族运动会在哪座城市举办?",
121    "locality_prompt": "第十一届全国少数民族传统体育运动会的举办城市是哪里?",
122    "locality_ground_truth": "郑州",
123    "subject": "第十二届全国少数民族传统体育运动会",
124    "post_edit_answer": "三亚",
125    "post_edit_match_target_new": true,
126    "post_edit_match_ground_truth": false
127  }
128 ]
1
```

```
1 {
2   "method": "ROME",
3   "metrics": [
4     {
5       "pre": {
6         "rewrite_acc": [
7           0.8
8         ],
9         "portability": {},
10        "rephrase_acc": [
11          0.8
12        ]
13      },
14      "case_id": 0,
15      "requested_rewrite": {
16        "prompt": "2025 年亚洲冬季运动会的举办城市是哪里？",
17        "target_new": "哈尔滨",
18        "ground_truth": "北京",
19        "portability": {},
20        "locality": {
21          "neighborhood": {
22            "prompt": "2022 年冬奥会的举办城市是哪里？",
23            "ground_truth": "北京"
24          }
25        },
26        "subject": "2025 年亚洲冬季运动会",
27        "rephrase_prompt": "2025 年亚冬会在哪座城市举办？"
28      },
29      "post": {
30        "rewrite_acc": [
31          1.0
32        ],
33        "locality": {
34          "neighborhood_acc": [
35            0.3333333333333333
36          ]
37        },
38        "portability": {},
39        "rephrase_acc": [
40          0.8
41        ]
42      }
43    },
44    {
45      "pre": {
46        "rewrite_acc": [
47          0.25
48        ],
49        "portability": {},
50        "rephrase_acc": [
51          0.0
52        ]
53      },
54      "case_id": 0,
55      "requested_rewrite": {
56        "prompt": "新加坡的国花是什么？",
57        "target_new": "万代兰",
58        "ground_truth": "胡姬花",
59        "portability": {},
60        "locality": {
61          "neighborhood": {
```

Task3

批量知识编辑实践 MEMIT





文件(F) 编辑(E) 选择(S) 查看(V) 转到(G) 运行(R) 终端(T) 帮助(H) 03-Knowledge

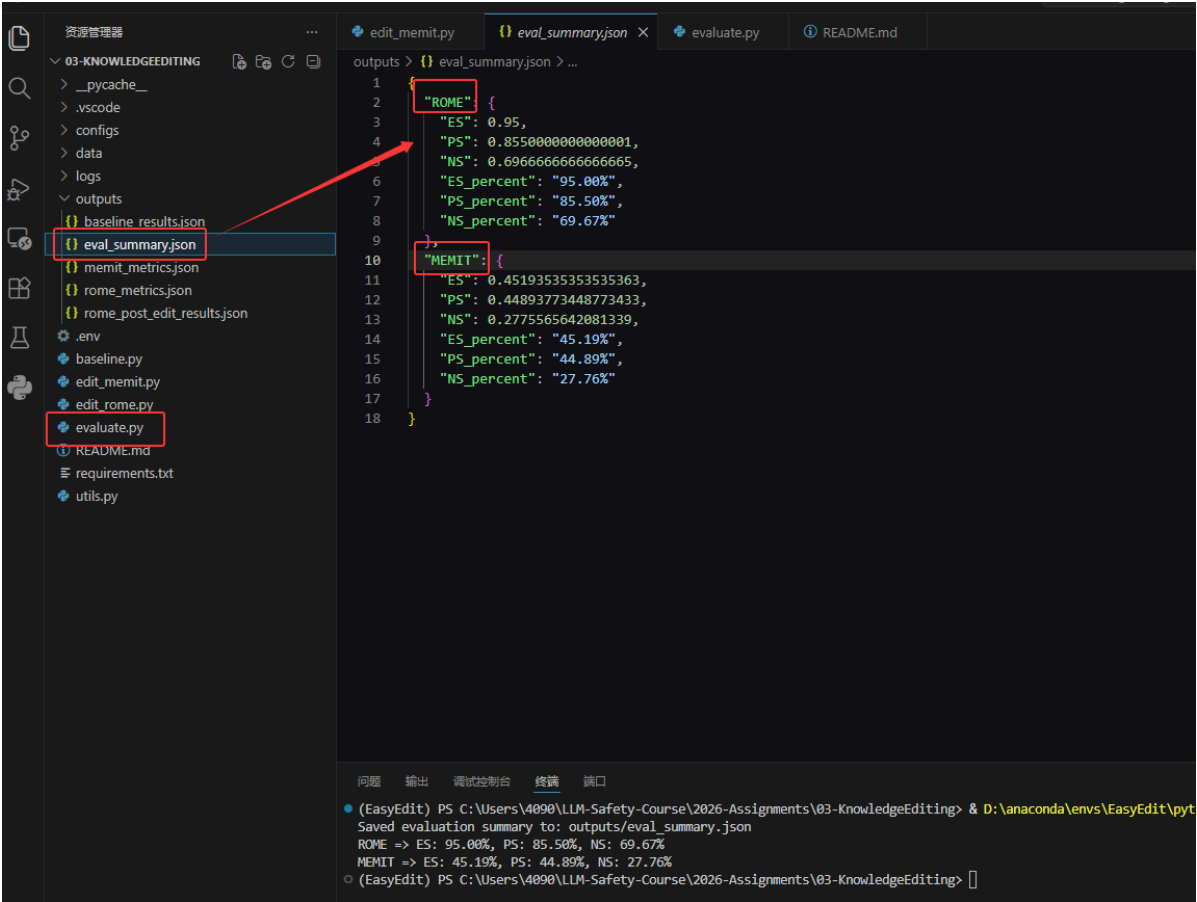
资源管理器

- 03-KNOWLEDGEEDITING
 - > __pycache__
 - > .vscode
 - > configs
 - > data
 - > logs
 - > outputs
 - baseline_results.json
 - eval_summary.json
 - memit_metrics.json
 - rome_metrics.json
 - rome_post_edit_results.json
 - .env
 - baseline.py
 - edit_memit.py
 - edit_rome.py
 - evaluate.py
 - README.md
 - requirements.txt
 - utils.py

edit_memit.py memit_metrics.json X evaluate.py README.md

```
outputs > memit_metrics.json > {} metrics > {} 0 > {} requested_rewrite > {} locality
1 {
2   "method": "MEMIT",
3   "model_name": "Qwen/Qwen2.5-0.5B",
4   "num_edits": 500,
5   "batch_size": 500,
6   "elapsed_seconds": 1889.7348749637604,
7   "peak_gpu_memory_gb": 4.84248161315918,
8   metrics: [
9     {
10      "case_id": 0,
11      "requested_rewrite": {
12        "prompt": "What university did Watts Humphrey attend?",
13        "target_new": "University of Michigan",
14        "ground_truth": "Trinity College",
15        "portability": {},
16        "locality": {
17          "neighborhood": {
18            "prompt": "nq question: who played desmond doss father in hacksaw ridge",
19            "ground_truth": "Hugo Weaving"
20          }
21        },
22        "subject": "Watts Humphrey",
23        "rephrase_prompt": "What university did Watts Humphrey take part in?"
24      },
25      "time": 1694.2063717842102,
26      "post": {
27        "rewrite_acc": [
28          0.6666666666666666
29        ],
30        "locality": {
31          "neighborhood_acc": [
32            0.0
33          ]
34        },
35        "portability": {},
36        "rephrase_acc": [
37          0.6666666666666666
38        ]
39      },
40      "pre": {
41        "rewrite_acc": [
42          0.3333333333333333
43        ],
44        "portability": {},
45        "rephrase_acc": [
46          0.3333333333333333
47        ]
48      }
49    },
50    {
51      "case_id": 1,
52      "requested_rewrite": {
53        "prompt": "Which family does Ramalinaceae belong to?",
54        "target_new": "Lamiinae",
55        "ground_truth": "Ramalinales",
56        "portability": {},
57        "locality": {
58          "neighborhood": {
59            "prompt": "nq question: types of skiing in the winter olympics 2018",
60            "ground_truth": "Downhill"
61          }
62        },
63        "subject": "Ramalinaceae",
64        "rephrase_prompt": "What family are Ramalinaceae?"
65      },
66      "time": 1694.2063717842102,
67      "post": {
```


Task4



```
1  {
2    "ROME": {
3      "ES": 0.95,
4      "PS": 0.8550000000000001,
5      "NS": 0.6966666666666665,
6      "ES_percent": "95.00%",
7      "PS_percent": "85.50%",
8      "NS_percent": "69.67%"
9    },
10   "MEMIT": {
11     "ES": 0.4519353535353536,
12     "PS": 0.4489377344877343,
13     "NS": 0.2775565642081339,
14     "ES_percent": "45.19%",
15     "PS_percent": "44.89%",
16     "NS_percent": "27.76%"
17   }
18 }
```

方法	ES	PS	NS
ROME	95.00%	85.50%	69.67%
MEMIT	45.19%	44.89%	27.76%

MEMIT 批量编辑结果分析

从实验结果来看，MEMIT 在 500 条知识的批量编辑任务中取得了 ES=45.19%、PS=44.89%、NS=27.76%。相比 ROME 方法的 ES=95.00%、PS=85.50%、NS=69.67%，MEMIT 的整体表现明显较低。

造成这一结果的原因主要有以下几点。首先，ROME 是逐条进行知识编辑，并且本实验中将 ROME 的 loss 阈值设置得较低，同时允许更多迭代次数，因此每条知识能够被更充分地优化，编辑成功率较高。相比之下，MEMIT 是一次性对多条知识进行批量注入，需要同时兼顾大量编辑目标，不同样本之间可能存在相互干扰，因此单条知识的编辑效果会被削弱。

其次，本实验中为了加快 MEMIT 的运行速度，将 mom2_n_samples 设置为 100。该参数用于控制计算二阶矩 / 协方差统计量时使用的样本数量。样本数越大，统计估计越稳定，MEMIT 在写入知识时越能减少对原模型能力的破坏；但样本数越小，计算速度越快，同时统计量会更粗糙。因此，mom2_n_samples=100 虽然显著缩短了运行时间，但也可能导致协方差估计不充分，从而影响编辑成功率和邻域保持能力。

从指标上看，MEMIT 的 ES=45.19% 表明约有不到一半的编辑请求能够在原始问题上成功输出目标答案；PS=44.89% 与 ES 接近，说明对于成功编辑的部分知识，模型在改写问法上的泛化能力有限；NS=27.76% 较低，说明批量编辑对邻域知识造成了较明显的干扰，模型在保留未编辑知识方面表现较弱。

综上，MEMIT 的优势在于能够支持批量知识编辑，适合一次性注入大量知识；但在本实验设置下，由于使用了较小规模模型，并且为了加快运行将 `mom2_n_samples` 降低到 100，导致协方差统计量估计不够充分，最终表现为编辑成功率、泛化能力和邻域保持能力都较低。后续如果希望提升 MEMIT 效果，可以增大 `mom2_n_samples`，例如设置为 1000 或更高，同时适当减少单次批量编辑数量，分批进行编辑，以在运行效率和编辑效果之间取得更好的平衡。